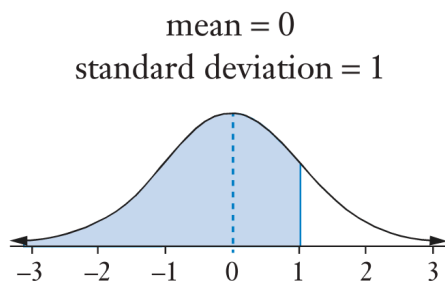


## Normal Distribution Examples

### Calculating Probabilities

#### Question 1.

Find  $P(X < 1)$  to 4 decimal places.

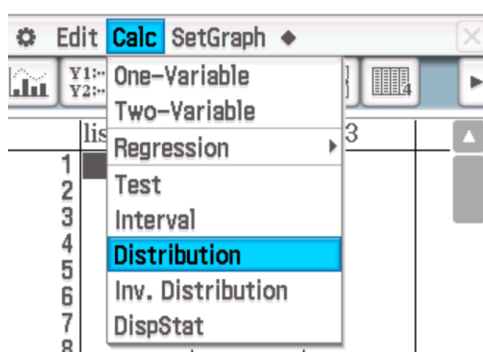


#### Using Classpad:

- 1) From Menu, click on the “Statistics” app



- 2) Click Calc > Distribution



- 3) Select “Normal CD” from the second drop down menu



- 4) Enter the lower boundary, upper boundary, mean and standard deviation then click "Next"

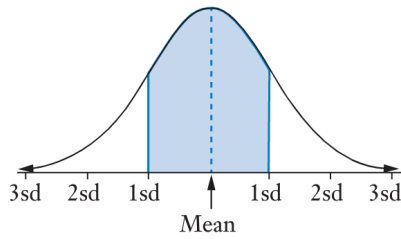
|          |                                 |
|----------|---------------------------------|
| Lower    | <input type="text" value="-∞"/> |
| Upper    | <input type="text" value="1"/>  |
| $\sigma$ | <input type="text" value="1"/>  |
| $\mu$    | <input type="text" value="0"/>  |

|                                               |                               |                                               |
|-----------------------------------------------|-------------------------------|-----------------------------------------------|
| <input type="button" value=" &lt;&lt; Back"/> | <input type="checkbox"/> Help | <input type="button" value=" Next &gt;&gt;"/> |
|-----------------------------------------------|-------------------------------|-----------------------------------------------|

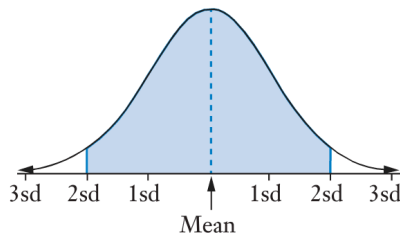
|          |                                        |
|----------|----------------------------------------|
| prob     | <input type="text" value="0.8413447"/> |
| z Low    | <input type="text" value="-∞"/>        |
| z Up     | <input type="text" value="1"/>         |
| $\sigma$ | <input type="text" value="1"/>         |
| $\mu$    | <input type="text" value="0"/>         |

Hence, when  $\mu = 0$  and  $\sigma = 1$ ,  $P(X < 1) = 0.8413$  (4 d.p.).

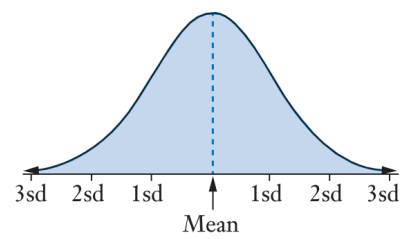
**By hand (using the 68.3, 95.4, 99.7% rule):**



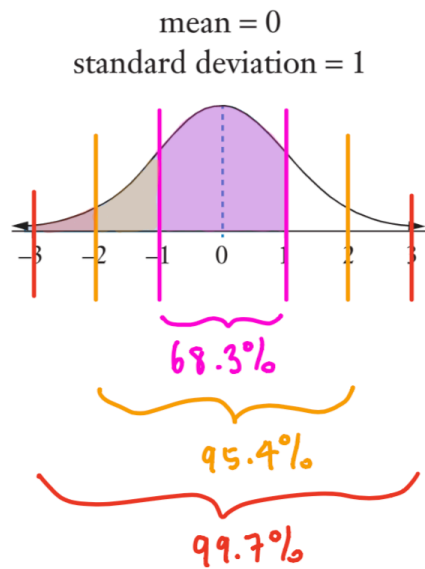
68.3% of area  
under curve shaded



95.4% of area  
under curve shaded



99.7% of area  
under curve shaded



$$P(X < 1) \approx 0.683 + \frac{0.954 - 0.683}{2} + \frac{0.997 - 0.954}{2}$$

$$\approx 0.84$$

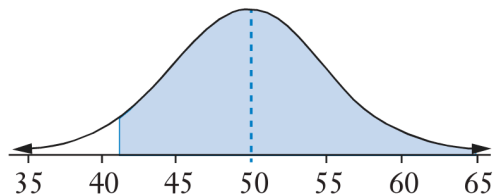
## Calculating boundaries given a probability:

### Question 2.

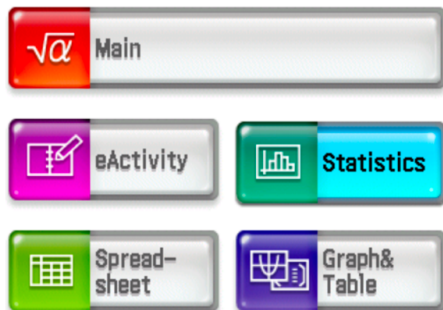
Find  $k$  given that  $P(X > k) = 0.9656$ .

mean = 50

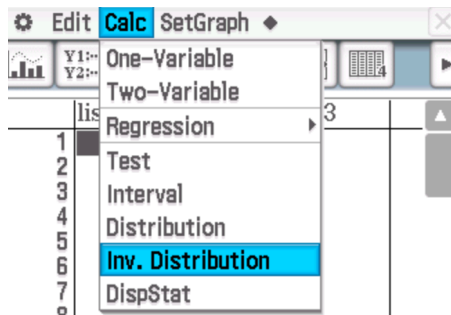
standard deviation = 5



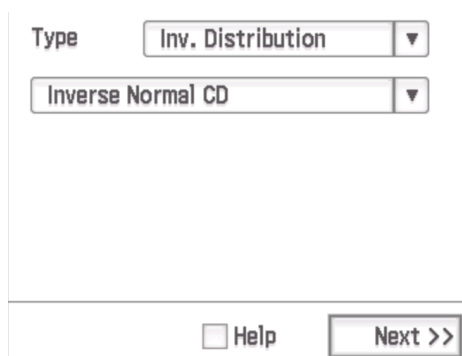
1) From Menu, click on the “Statistics” app



2) Click Calc > Inv. Distribution



3) Make sure “Inverse Normal CD” is selected then click “Next”



- 4) For this question, the probability given includes the **right tail** of the graph so change the tail setting to “Right”, enter the probability, mean, and standard deviation then click “Next”.

Tail setting **Right** ▼

prob 0.9656

$\sigma$  5

$\mu$  50

<< Back ☐ Help Next >>

$x_1 \text{InvN}$  40.901346

prob 0.9656

$\sigma$  5

$\mu$  50

Hence,  $k = 40.9$  (1 d.p.).

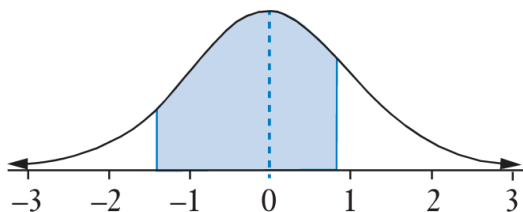
i.e. When  $\mu = 50$  and  $\sigma = 5$ ,  $P(X > \mathbf{40.9}) = 0.9656$ .

### Question 3.

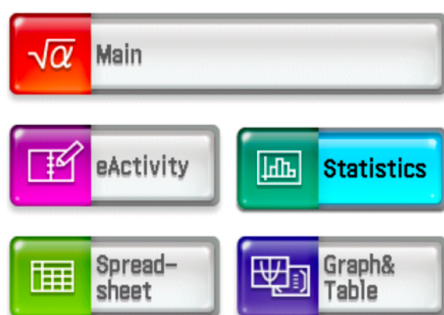
Find  $k$  given that  $P(-1.4 < X < k) = 0.7215$ .

mean = 0

standard deviation = 1



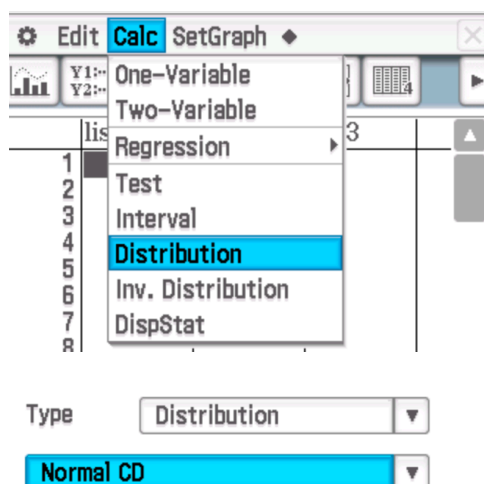
- 1) From Menu, click on the “Statistics” app



- 2) For this question, we cannot use the tail setting “Centre” because the shaded area isn’t centred about the mean. We also cannot use “Left” or “Right” because the area shaded includes neither the left nor right tail.

We can however create a “left tail” by calculating  $P(X < -1.4)$  then adding it to the shaded area.

In Statistics, click Calc > Distribution, select “Normal CD” from the second drop down menu, click “Next”, enter the lower boundary, upper boundary



Lower  $-\infty$   
 Upper  $-1.4$   
 $\sigma$  1  
 $\mu$  0

<< Back ☐ Help Next >>

prob 0.0807567  
 z Low  $-\infty$   
 z Up  $-1.4$   
 $\sigma$  1  
 $\mu$  0

<< Back ☐ Help

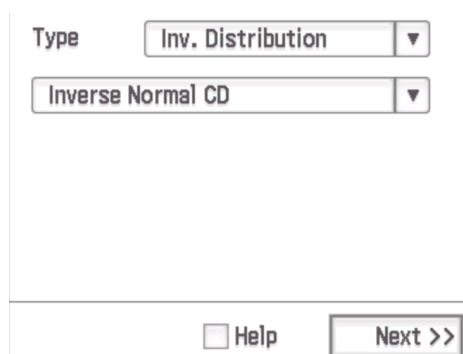
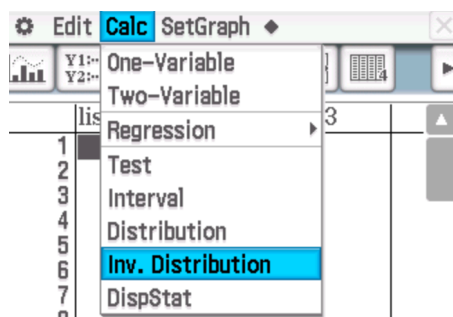
Hence  $P(X < -1.4) = 0.0807567$

$$P(X < k) = P(X < -1.4) + P(-1.4 < X < k)$$

$$P(X < k) = 0.0807567 + 0.7215$$

$$P(X < k) = 0.8022567$$

3) Click Calc > Inv. Distribution, select “Inverse Normal CD” then click “Next”



Enter the tail setting, probability, mean and standard deviation then click “Next”. In this case, we created a “left tail” and the new probability is 0.8022567.

Tail setting

Left

▲

prob

0.8022567

$\sigma$

1

$\mu$

0

▼

<< Back

☐ Help

Next >>

$x_1 \text{InvN}$

0.8497095

prob

0.8022567

$\sigma$

1

$\mu$

0

<< Back

☐ Help

Hence,  $k = 0.85$  (2 d.p.).

i.e. When  $\mu = 0$  and  $\sigma = 1$ ,  $P(-1.4 < X < \mathbf{0.85}) = 0.7215$ .